

hip angular
motion

hip pitch

$r_{hk} \approx 0.4 \text{ m}$

knee pitch

a_{O_h} carried motion

$\alpha_h \times r_{hk}$

$\omega_h \times (\omega_h \times r_{hk})$

The knee origin is moving.
A single-joint knee model
misses these parent-link
acceleration inputs.

